Task 2: Word-Level Language Model

Introduction: This task involved implementing a word-level language model trained on *Poirot Investigates* by Agatha Christie. The objective was to predict the next word in a sequence and generate at least 20 subsequent words given an input prompt ranging from a single word to a paragraph.

The dataset was preprocessed by removing Project Gutenberg metadata, converting text to lowercase, and tokenizing at word level. The final corpus contained approximately 52,543 words with a vocabulary of 9,048 unique tokens. Training samples were generated using a sliding window approach with a sequence length of 20 words, where the model predicts the next word given the previous 20 words. A sequence length of 20 was selected to balance contextual learning and computational efficiency.

A two-layer **LSTM model** was implemented using PyTorch. The architecture consists of an embedding layer (128 dimensions), two LSTM layers (hidden size = 256), dropout (0.2) for regularization, and a fully connected output layer. The model was trained for 20 epochs using the Adam optimizer (learning rate = 0.001) and CrossEntropyLoss with a batch size of 64. Training loss decreased from 7.10 to 1.25, demonstrating effective learning of statistical word dependencies. LSTM was chosen instead of a vanilla RNN to mitigate vanishing gradient issues and instead of a Transformer due to the relatively small dataset size (~52k words), as transformer models typically require much larger corpora.

During inference, the model accepts arbitrary user input and generates 80 words using temperature sampling (≥20 as required). The generated text exhibits grammatical sentence structure and captures surface-level stylistic patterns of Christie's writing. However, while local coherence within short spans is maintained, the model occasionally repeats phrases and struggles with long-range semantic consistency due to its limited 20-word context window and sequential hidden-state processing.

In contrast, ChatGPT is based on a transformer architecture trained on billions of tokens and utilizes self-attention mechanisms to model long-range dependencies. Unlike LSTMs, transformers attend all tokens within the context window simultaneously, enabling stronger global coherence and thematic consistency. Consequently, ChatGPT produces more semantically consistent and contextually aware of text, whereas the LSTM model primarily captures statistical word transitions and stylistic features. This comparison highlights the impact of architectural design, attention mechanisms, and dataset scales in modern language modelling.